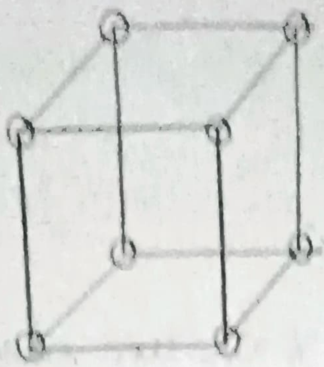
Simple Cubic Structure



This shows simple cubic structure. Each atom at corner surrounded by 8 neighbour atoms.

i) Number of atoms per unit cell

It has 8 corners, each corners has one in atom. Each atom shared by 8 neighbour atoms. So each atom will give its $\frac{1}{8}$th part to unit cell.

Total No. of atoms in unit cell

$= \frac{1}{8} \times$ total No. of corner atoms

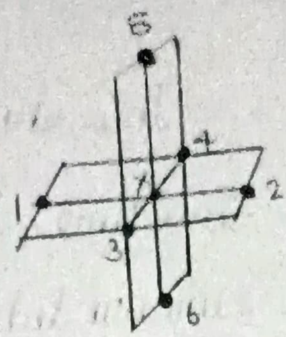
$= \frac{1}{8} \times 8$

$= 1$

Total No. of atoms per unit cell $= 1$

It is a primitive cell.

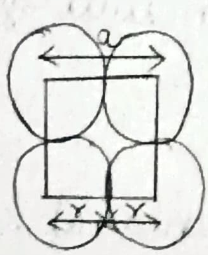
Coordination Number



If we consider x as an atom, it is surrounded 6 other atoms

Coordination No = 6

Atomic Radius



Consider a face of unit cell. Let a be the side and r be the radius of unit cell

$$2r = a$$

$$r = \frac{a}{2}$$

Packing factor

No. of atoms per unit cell = 1

Volume of one atom = $\frac{4}{3}\pi r^3$

Volume of atoms in unit cell, $v = 1 \times \frac{4}{3}\pi r^3$

Volume of the unit cell, $V = a^3$

Packing factor $P.F = \frac{v}{V}$

$= \frac{4}{3}\pi r^3 \times \frac{1}{a^3}$

$= \frac{4}{3}\pi \left(\frac{a}{2}\right)^3 \times \frac{1}{a^3}$

$= \frac{4}{3}\pi \frac{1}{8}$

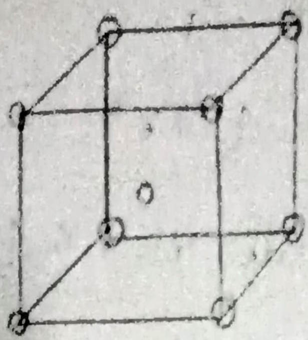
$= \frac{\pi}{6}$

$= 0.52$

$= 52\%$

Eg: Polonium

Body Centered Cubic Structure



It 8 corners, each corner has one atom and one more atom present at centre of the body.

No. of atoms per unit cell:

It has 8 corner atoms, each corner has one atom, each atom shared by 8 neighbour atoms. So each atom will give its $\frac{1}{8}$ Th part to one unit cell.

Total No. of corner atoms in one unit cell

$= \frac{1}{8} \times$ total No. of corner atoms

$= \frac{1}{8} \times 8$

$= 1$

No. of body centered atoms in one unit cell $= 1$

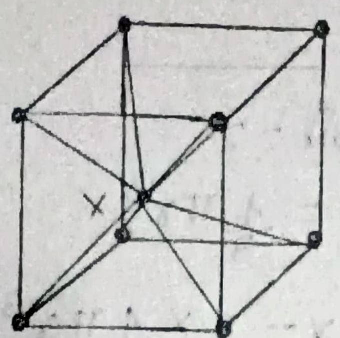
Total No. of atoms in one cell

$=$ Total No. of corner atoms + No. of bc atoms

$= 1 + 1$

$= 2$

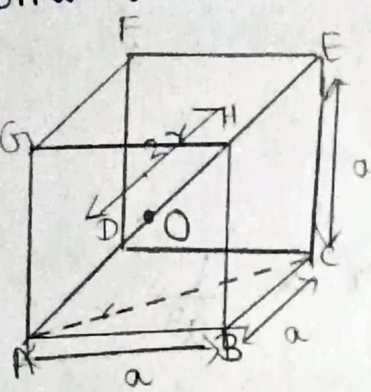
Coordination Number



If X is body centered atom surrounded by 8 corner atoms

Coordination No. = 8

Atomic Radius



Let a be the area and r be the radius of the unit cell.

From figure

$$AC^2 = AB^2 + BC^2$$

$$= a^2 + a^2$$

$$AC = \sqrt{2}a$$

From figure,

$$AE^2 = AB^2 + BC^2 + CE^2$$

$$(4r)^2 = a^2 + a^2 + a^2$$

$$16r^2 = 3a^2$$

$$r^2 = \frac{3a^2}{16}$$

$$r = \frac{\sqrt{3}a}{4}$$

$$a = \frac{4r}{\sqrt{3}}$$

Packing factor

No. of atoms per unit cell = 2

Volume of one atom $= \frac{4}{3}\pi r^3$

Volume of two atoms $v = 2 \times \frac{4}{3}\pi r^3$

Volume of unit cell $V = a \times a \times a = a^3$

$$P.F = \frac{v}{V}$$

$$= 2 \times \frac{4}{3}\pi r^3 \times \frac{1}{a^3}$$

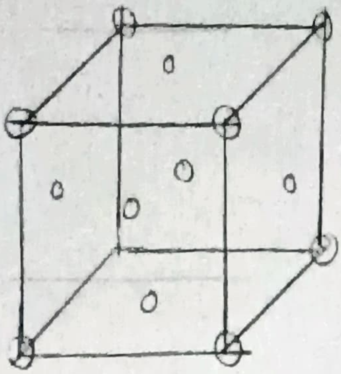
$$= 2 \times \frac{4\pi}{\cancel{3}} \frac{\cancel{3}\sqrt{3}a^3}{4 \times 16\,8} \times \frac{1}{\cancel{a^3}}$$

$$= \frac{\sqrt{3}\pi}{8}$$

$$= 68\%$$

Eg: Chromium.

Face centered cubic structure



It has 8 corners and each has one atom, it has 6 atoms at face and one atom at centre

No. of atoms per unit cell

It has 8 corner atoms, each corner has one atom is shared by 8 atoms.

No. of atoms in unit cell due to corner atoms $= \frac{1}{8} \times 8 = 1$

Also it has 6 face centered atoms shared by 2 unit cells

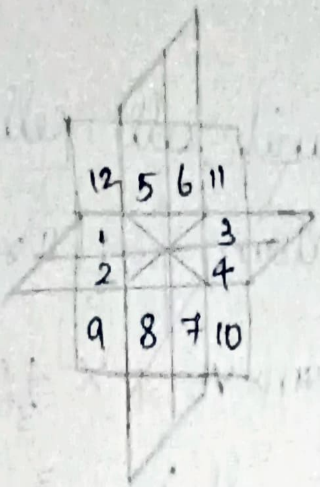
No. of face centered atoms $= \frac{1}{2} \times 6 = 3$

Total No. of atoms in unit cell

$= 1 + 3$

$= 4$

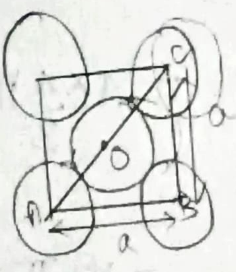
Coordination Number



FCC has 8 atoms at corner, 6 atoms at centers of six faces. If we consider corner atom as X, it is surrounded by 12

∴ Coordination No = 12

Atomic Radius



From ΔABC,

$$AC^2 = AB^2 + BC^2$$

$$(2r)^2 = a^2 + a^2$$

$$4r^2 = 2a^2$$

$$r^2 = \frac{2a^2}{4}$$

$$r = \frac{\sqrt{2}a}{4}$$

$$a = \frac{4r}{\sqrt{2}}$$

Packing factor

No. of atoms per unit cell cell $= 4$

Volume of one atom $= \frac{4}{3}\pi r^3$

Volume of 4 atoms $v = 4 \times \frac{4}{3}\pi r^3$

Volume of unit cell $V = a^3$

Packing factor $PF = \frac{v}{V}$

$$= 4 \times \frac{4}{3}\pi r^3 \times \frac{1}{a^3}$$

$$= 4 \times \frac{4}{3}\pi \times \frac{2\sqrt{2}\,a^3}{4 \times 4 \times 4} \times \frac{1}{a^3}$$

$$= \frac{\sqrt{2}\,\pi}{6}$$

$$= 0.74$$

$$= 74\%$$

Eg: Gold.